

VisionAid: A Multimodal Assistive Application Supporting Safe Road Navigation for Visually Impaired People in Bangladesh

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Abstract

Navigating urban environments presents significant challenges for visually impaired individuals, particularly when crossing roads in unstructured traffic conditions. Traditional mobility aids, such as white canes, offer limited situational awareness of dynamic obstacles like moving vehicles. To address this gap, we present VisionAid, a mobile application that utilizes real-time obstacle detection, traffic signals, and pedestrian crossings. The system provides multimodal feedback using context-aware natural language announcements for scene description and intensity-based haptic patterns—such as short pulses for warnings and continuous vibrations for immediate hazards—enabling users to interpret their surroundings effectively. We evaluated the system in real-world scenarios in Bangladesh, demonstrating that our novel vehicle occupancy algorithm effectively identifies road hazards with low latency on consumer-grade smartphones.

CCS Concepts

• **Human-centered computing** → Accessibility technologies; Ubiquitous and mobile computing systems; • **Computing methodologies** → Artificial intelligence.

Keywords

Assistive Technology, Computer Vision, Mobile HCI, Haptic Feedback, Android

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1 Introduction

Independent urban mobility remains a critical challenge for the estimated 2.2 billion [16] visually impaired individuals worldwide. The most perilous task is navigating dynamic road environments, where oncoming traffic poses an immediate threat that cannot be detected by traditional tactile aids.

While traditional aids like the white cane are effective for identifying near-field static obstacles, they fail to provide the long-range situational awareness necessary to judge traffic flow, creating a significant "information gap." To bridge this gap, prior research in HCI has explored diverse technological interventions. Hardware-centric systems, such as the WOAD [3] and ALVU [5], employ wearable sensors to map environmental depth and deliver haptic feedback, while infrastructure-based solutions like NavCog [1] utilize Bluetooth beacons for precise indoor guidance. On the mobile front, recent work has focused on software-based assistance: DeepNAVI [7] leverages deep learning to narrate scene context, PathFinder [8] introduces map-less navigation models, and WatchOut [13] utilizes advanced sonification to convey real-time obstacle characteristics. However, while these innovations have significantly advanced accessibility, they often prioritize structured environments or general navigation over the immediate, safety-critical analysis required for chaotic street crossings in countries such as Bangladesh.

To address this, we present VisionAid, a smartphone-based navigation assistant that leverages real-time computer vision to democratize safe mobility. Our primary contribution is the development of a resource-efficient Vehicle Occupancy Heuristic—a safety logic designed specifically for chaotic traffic. In busy streets of Bangladesh, standard object detection models generate excessive noise by alerting users to every vehicle in view. Our algorithm, conversely, analyzes the spatial dominance of vehicles within the camera frame to distinguish between distant, safe traffic and immediate, high-risk hazards. By optimizing lightweight models (YOLOv11 via TensorFlow Lite), we enable this safety-critical processing to run locally on commodity smartphones with low latency (120ms), eliminating the need for expensive, specialized hardware.

In rapidly developing cities in South Asia, especially in Bangladesh, traffic is highly unstructured; lane discipline is rare, signal violations are frequent, and pedestrian infrastructure is inconsistent. Western-centric navigation aids, which often rely on structured intersections and predictable traffic flow, are unreliable in such settings. VisionAid is engineered to handle this complexity, providing context-aware audio and haptic feedback that allows users to navigate unpredictable environments safely and independently.

2 Related Work

Recent research in assistive navigation spans diverse modalities, ranging from specialized wearables to pure software solutions. We categorize these developments into hardware-centric aids, infrastructure-based systems, and smartphone-based assistants, positioning VisionAid within this landscape.



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2.1 Wearable & Hardware-Centric Aids

A significant body of work augments traditional mobility tools with advanced sensors. Systems like ALVU [5] and WOAD [3] utilize Time-of-Flight sensors and depth-aided video compression to provide high-fidelity environmental mapping. The integration of Generative AI and advanced sensing represents the frontier of this domain; for instance, LLM-Glasses [15] leverage multimodal LLMs for directional reasoning, while specialized attachments like the smartphone-mounted LiDAR in Corridor-Walker [10] construct accurate 2D occupancy grids for indoor intersection recognition. Similarly, electronic cane prototypes [11, 14] embed ultrasonic sensors directly into the white cane. However, despite their robustness, these hardware-centric solutions present high barriers to adoption in the Global South due to cost, bulk, and maintenance requirements unsuitable for resource-constrained environments.

2.2 Infrastructure & Sensor-Based Solutions

To reduce the burden on the user, some systems rely on environmental instrumentation. NavCog [1], for example, uses Bluetooth Low Energy (BLE) beacons for precise turn-by-turn indoor navigation. While effective in controlled settings, infrastructure-based approaches face severe scalability challenges in outdoor urban environments. In the sprawling, unstructured street networks of cities, maintaining a dense sensor grid is practically unfeasible, necessitating a self-contained solution.

2.3 Smartphone-Based Assistants

The ubiquity of smartphones has shifted focus toward accessible, software-first solutions. DeepNAVI [7] pioneered the use of deep learning for scene narration, serving as a foundational inspiration for our work. Recent work in HCI and CHI have further diversified this space: PathFinder [8] introduced a map-less navigation model to reduce cognitive load. Other applications address highly specific situational needs, such as LineChaser [9], which utilizes pedestrian detection and depth sensors to help users independently navigate and stand in queues. Furthermore, systems like WatchOut [13] focus on conveying real-time characteristics of nearby hazards through advanced obstacle sonification techniques without requiring dedicated wearables.

However, existing smartphone solutions often prioritize recognition or localization over immediate safety. Commercial apps like Microsoft's Seeing AI [12] excel at static object identification, and specialized tools like Oko [2] focus on standardized traffic signals. These Western-centric paradigms often fail in Bangladesh's chaotic traffic, where lane discipline is non-existent and drivers frequently violate signals. Unlike prior works focused on scene description, we introduce a resource-efficient "vehicle occupancy" heuristic to assess collision risks in real-time. By combining this logic with intensity-based haptic patterns—building on findings by Jiang et al. [4] and Khushro et al. [6]—VisionAid delivers a holistic safety tool resilient to the unpredictability of unstructured urban environments.

3 System Design

VisionAid is designed as a real-time Android application. The application's core architecture consists of three main stages: Frame Preprocessing, Core Inference, and Feedback Logic (see Fig. 1).

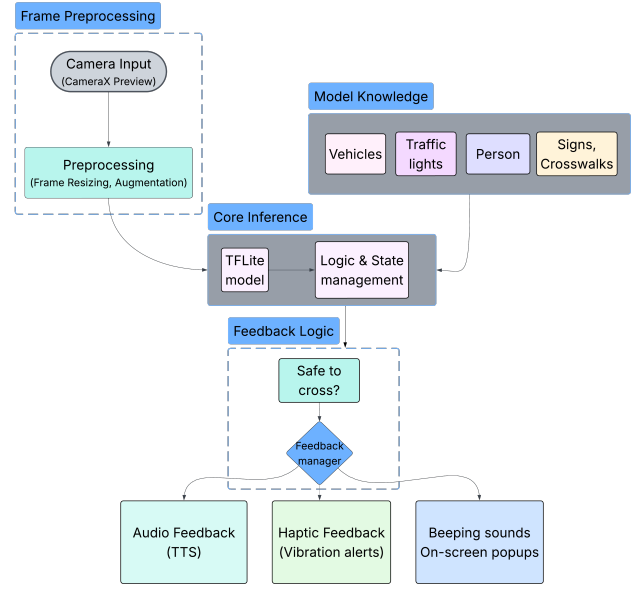


Figure 1: VisionAid System Architecture detailing the three-stage pipeline: Frame Preprocessing of camera input, Core Inference via a unified YOLOv11 model, and multimodal Feedback Logic.

3.1 Frame Preprocessing

The system initiates by capturing a high-frequency video stream using the Android CameraX API (see Fig. 2), which ensures broad compatibility across different device hardware while minimizing camera initialization latency. To maintain real-time performance, each captured frame undergoes immediate preprocessing. The system resizes and rotates the raw bitmap data to match the specific input tensor resolution required by the neural network, ensuring the data is normalized and noise-free before inference.

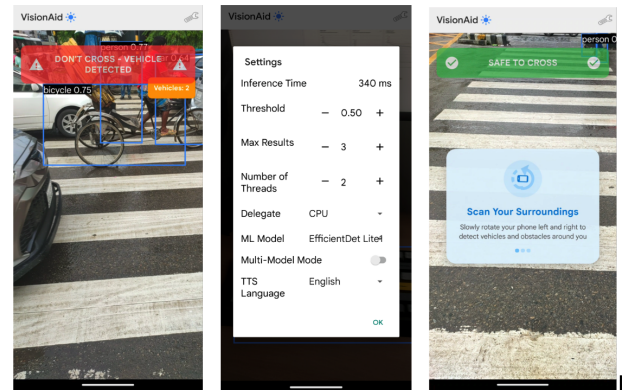


Figure 2: VisionAid System Overview: (Left), Main Interface with live detection; (Middle), Settings Menu for thresholds and feedback; (Right), Field Test demonstrating real-time vehicle recognition and safety alerts.

3.2 Core Inference

The core intelligence of VisionAid relies on a quantized YOLOv11 model running on the TensorFlow Lite Interpreter. Initially, the system utilized separate models for different tasks (traffic lights vs. vehicles), which resulted in high latency (~400ms). By consolidating these into a single unified model trained on a custom dataset of 9,500 images—covering local classes like rickshaws, buses, and potholes—we reduced inference time to approximately 120ms on mid-range devices.

Within this inference loop, we implement a custom Vehicle Occupancy Heuristic to handle chaotic traffic. Rather than simply detecting the presence of a car, which creates excessive noise in busy streets, the system calculates a "threat level" based on frame occupancy. A "Vehicle Detected" (Don't Cross) state is triggered only if:

$$\sum A_{vehicle} > 0.6 \times A_{frame} \quad (1)$$

where $A_{vehicle}$ is the cumulative area of vehicle bounding boxes and A_{frame} is the total screen area. To prevent transient noise—such as rapid camera movement or a vehicle briefly passing the periphery—from triggering false alarms, this condition acts as a temporal filter. Specifically, the spatial dominance defined in Equation 1 must persist across consecutive frames for approximately one second before a 'Don't Cross' hazard alert is executed. This ensures the system only alerts users to vehicles that are genuinely blocking their path or approaching dangerously close.

3.3 Feedback Logic

The final stage translates the inference results into intuitive, multimodal alerts designed for visually impaired users. The Feedback Manager ensures reliability through redundancy:

- **Audio Announcements:** The Android Text-to-Speech (TTS) engine provides descriptive context, such as "Vehicles present, be careful" or "Safe to cross".
- **Haptic Feedback:** To accommodate noisy outdoor environments where audio might be drowned out, the system uses the Android Vibrator service to deliver distinct intensity-based patterns. Short pulses indicate static warnings (e.g., a stop sign), while longer, continuous vibrations signal immediate hazards.
- **Auditory Beeps:** Tone generator plays distinct beep patterns corresponding to the density of traffic, allowing users to gauge danger levels immediately without waiting for full speech sentences to finish.

4 System Evaluation

4.1 Model Performance

We initially developed specialized single-task models (e.g., YOLOv8n for crosswalks) and implemented a feature allowing the simultaneous execution of multiple models (e.g., combining traffic light and vehicle detection). While this multi-model mode remains accessible in the application settings for experimental use, utilizing it in real-time scenarios created a significant performance bottleneck, driving inference latency up to 300–400ms per frame. This delay was unacceptable for a safety-critical application requiring immediate feedback.

To resolve this, we consolidated the detection tasks into a single Unified YOLOv11 model. While this resulted in a slight trade-off in raw precision compared to the specialized models, it drastically reduced the computational overhead. Table 1 shows a comparison for our three test implementations where the unified model achieved an inference time of approximately 120ms—a 70% reduction—enabling the application to run smoothly at 8–10 FPS on mid-range devices.

Table 1: Model Performance Comparison

Model Type	Precision	Recall	Inference Time
Crosswalk (YOLOv8n)	94.7%	87.8%	50ms
Traffic Light (YOLOv8)	93.7%	80.7%	45ms
Unified YOLOv11	82.4%	68.5%	120ms

4.2 Field Testing

We conducted a pilot field study with 6 participants with multiple road crossing attempts (aged 18-24, non-vision-impaired) to evaluate the system's efficacy in real-world conditions. Testing took place across four distinct urban environments, ranging from chaotic high-traffic intersections in busy commercial area to semi-controlled walkways within a university campus (see Fig. 3).



Figure 3: Field testing with participants at various urban locations, including high-traffic intersections and campus walkways.

The unified model maintained a stable frame rate of 8–10 FPS, which was sufficient for normal walking speeds. Battery consumption was approximately 15% during a continuous 16-minute stress test (with the screen active and ML inference running). Additionally, the occupancy heuristic successfully filtered out distant traffic noise, triggering alerts only when vehicles breached the safety threshold (occupying > 60% of the frame). While this level of consumption is viable for short-to-medium commutes, future iterations might optimize power draw by utilizing the smartphone's accelerometer to dynamically lower the camera frame rate when the user is standing still.

5 Discussion

Deploying computer vision solutions in unstructured traffic environments like Bangladesh presents unique challenges compared to the structured roadways often assumed by Western-centric navigation aids. In a crowded city, we observed that relying solely on standard cues—such as traffic light detection or simple object counting—was insufficient for user safety. Field tests revealed frequent signal violations, such as vehicles continuing to move during green pedestrian lights, rendering these static indicators unreliable.

To address this, our system prioritizes a "threat-first" approach over simple object recognition. The heuristic-based "vehicle occupancy" metric proved effective by filtering out transient noise and focusing on immediate hazards. By requiring vehicles to sustain a specific frame occupancy threshold for approximately one second, the algorithm successfully distinguishes between safe, distant traffic and dangerous, approaching vehicles, significantly reducing false positives compared to raw detection streams.

From a systems perspective, VisionAid features a highly modular architecture that allows any custom-trained ML model to be easily integrated. This flexibility facilitated the transition from multiple specialized models to a single unified YOLOv11 architecture, a critical optimization that addressed early performance bottlenecks by reducing inference latency from ~400ms to approximately 120ms. This 70% reduction in processing time enabled the application to maintain a consistent 8–10 FPS on mid-range devices, mitigating overheating and lag issues. While the unified YOLOv11 model exhibits a lower overall recall (68.5%) compared to specialized models, this metric equally penalizes missed detections of distant, occluded vehicles. In the context of immediate pedestrian safety, this drop in recall is mitigated by the Vehicle Occupancy Heuristic. The heuristic acts as a fail-safe; it does not require perfect bounding-box recall for background traffic, but rather ensures high confidence in detecting vehicles that breach the critical $> 60\%$ spatial threshold, effectively minimizing dangerous false negatives in the immediate crossing zone. Crucially, this modular foundation ensures the inference engine can be effortlessly upgraded to improve performance in the future as needed.

Despite these advancements, the current system operates within certain constraints. While continuous inference is efficient, it remains power-intensive; future iterations will optimize battery consumption by utilizing the smartphone's accelerometer to dynamically lower the frame rate when the user is stationary. Furthermore, as VisionAid relies entirely on optical camera input, its reliability degrades in low-light conditions (such as nighttime) or adverse weather like heavy rain or fog. Additionally, without hardware depth sensors like LiDAR, distance estimation depends on 2D bounding box heuristics, which may lack precision when detecting objects at oblique angles or in highly crowded scenes—limitations we intend to address in future work.

6 Conclusion

Providing safe urban navigation for the visually impaired does not necessitate expensive, specialized hardware. This work presents a software-first approach that transforms commodity smartphones into reliable assistive tools capable of navigating chaotic urban traffic. By optimizing deep learning models for edge-level execution and implementing a novel Vehicle Occupancy Heuristic, we achieved a high-performance system that delivers safety-critical alerts in real-time. Our results highlight the potential for accessible, mobile-centric solutions to bridge the technological gap for visually impaired individuals in developing regions.

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References

- [1] Dragan Ahmetovic, Cole Gleason, Chengxiong Ruan, Kris Kitani, Hironobu Takagi, and Chieko Asakawa. 2016. NavCog: a navigational cognitive assistant for the blind. In *Proceedings of the 18th International Conference on Human-Computer Interaction with Mobile Devices and Services* (Florence, Italy) (*MobileHCI '16*). Association for Computing Machinery, New York, NY, USA, 90–99. doi:10.1145/2935334.2935361
- [2] Gus Alexiou. 2023. OKO App Leverages AI To Help Blind Pedestrians Recognize Traffic Signals. *Forbes.com*. <https://www.forbes.com/sites/gusalexioiu/2023/08/10/oko-app-deploys-ai-to-make-crossing-the-street-safer-for-blind-pedestrians/> Accessed: 2026-01-19.
- [3] Yun Gao, Dan Wu, Jie Song, Xueyi Zhang, Bangbang Hou, Hengfa Liu, Junqi Liao, and Liang Zhou. 2025. A wearable obstacle avoidance device for visually impaired individuals with cross-modal learning. *Nature Communications* 16, 1 (2025), 2857. doi:10.1038/s41467-025-58085-x
- [4] Chutian Jiang, Emily Kuang, and Mingming Fan. 2025. How Can Haptic Feedback Assist People with Blind and Low Vision (BLV): A Systematic Literature Review. *ACM Trans. Access. Comput.* 18, 1, Article 2 (Feb. 2025), 57 pages. doi:10.1145/3711931
- [5] Robert K. Katzschmann, Brandon Araki, and Daniela Rus. 2018. Safe Local Navigation for Visually Impaired Users With a Time-of-Flight and Haptic Feedback Device. *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 26, 3 (March 2018), 583–593. doi:10.1109/TNSRE.2018.2800665
- [6] Shah Khuroo, Babar Shah, Inayat Khan, and Sumayya Rahman. 2022. Haptic Feedback to Assist Blind People in Indoor Environment Using Vibration Patterns. *Sensors* 22, 1 (2022). doi:10.3390/s22010361
- [7] Bineeth Kuriakose, Raju Shrestha, and Frode Eika Sandnes. 2023. DeepNAVI: A deep learning based smartphone navigation assistant for people with visual impairments. *Expert Syst. Appl.* 212, C (Feb. 2023), 16 pages. doi:10.1016/j.eswa.2022.118720
- [8] Masaki Kuribayashi, Tatsuya Ishihara, Daisuke Sato, Jayakorn Vongkulbhisal, Karnik Ram, Seita Kayukawa, Hironobu Takagi, Shigeo Morishima, and Chieko Asakawa. 2023. PathFinder: Designing a Map-less Navigation System for Blind People in Unfamiliar Buildings. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (*CHI '23*). Association for Computing Machinery, New York, NY, USA, Article 41, 16 pages. doi:10.1145/3544548.3580687
- [9] Masaki Kuribayashi, Seita Kayukawa, Hironobu Takagi, Chieko Asakawa, and Shigeo Morishima. 2021. LineChaser: A Smartphone-Based Navigation System for Blind People to Stand in Lines. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (*CHI '21*). Association for Computing Machinery, New York, NY, USA, Article 33, 13 pages. doi:10.1145/3411764.3445451
- [10] Masaki Kuribayashi, Seita Kayukawa, Jayakorn Vongkulbhisal, Chieko Asakawa, Daisuke Sato, Hironobu Takagi, and Shigeo Morishima. 2022. Corridor-Walker: Mobile Indoor Walking Assistance for Blind People to Avoid Obstacles and Recognize Intersections. *Proc. ACM Hum.-Comput. Interact.* 6, MHCI, Article 179 (Sept. 2022), 22 pages. doi:10.1145/3546714
- [11] T Lavanya Narayani., M Sivapalanirajan., B Keerthika., M Ananthi., and M Arunarani. 2021. Design of Smart Cane with integrated camera module for visually impaired people. In *2021 International Conference on Artificial Intelligence and Smart Systems (ICAIS)*. 999–1004. doi:10.1109/ICAIS50930.2021.9395840
- [12] Microsoft. 2017. Seeing AI: Talking camera for the blind. <https://www.microsoft.com/en-us/ai/seeing-ai> Accessed: 2026-01-19.
- [13] Giorgio Presti, Dragan Ahmetovic, Mattia Ducci, Cristian Bernareggi, Luca Ludovico, Adriano Barattè, Federico Avanzini, and Sergio Mascetti. 2019. WatchOut: Obstacle Sonification for People with Visual Impairment or Blindness. In *Proceedings of the 21st International ACM SIGACCESS Conference on Computers and Accessibility* (Pittsburgh, PA, USA) (*ASSETS '19*). Association for Computing Machinery, New York, NY, USA, 402–413. doi:10.1145/3308561.3353779
- [14] Caio H. M. T., Amanda A. R., Ana Luiza F. d. A. C., and Vagner R. d. S. 2023. Wearable Haptic Device as Mobility Aid for Blind People: Electronic Cane - Wearable Device for Mobility of Blind People. *JOJ Ophthalmology* 9, 3 (2023), 555765. doi:10.19080/JOJO.2023.09.555765
- [15] Issatay Tokmurziyev, Miguel Altamirano Cabrera, Muhammad Haris Khan, Yara Mahmoud, and Dzmitry Tsetserukou. 2026. LLM-Glasses: GenAI-driven Glasses with Haptic Feedback for Navigation of Visually Impaired People. arXiv:2503.16475 [cs.HC] <https://arxiv.org/abs/2503.16475>
- [16] World Health Organization. 2023. Blindness and vision impairment. <https://www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment> Accessed: 2026-01-19.